L5.4: What is a linear mapping - Part 2

Transcript Language

English

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What is a linear mapping Part 2

Let us move on to a slightly more elaborate example. So, we will use the same data that

we have before, we have shop A. But now in this locality Malgudi, on this town Malgudi, we have the original shop A, and we have two other shops B and C as well, and the prices

of rice, dal and oil in each of the shops is given in this table.

So, in Shop A we already have seen that it is 45 kgs, 45 rupees per kg for rice, 125

rupees per kg for a dal and I think 150 rupees per liter for oil. And for Shop B we have

40, 120 and 170 respectively and Shop C we have 50, 130 and 160 respectively.

Now, someone asked the caterer why do you always buy from shop A? So, is there a good

way of deciding, which shop to buy your groceries from? So, in order to do this, what do we

have to do, we have to check out the cost for buying X1 kgs of rice, X2 kgs of dal and

X3 liters of oil from each of the shops, then compare them and see which one is less, and

then we would go there. Of course, let us assume that all of them have the same quality

and there is no difference as such.

So, let us do that. So, we have seen already that the expression for shop A is a linear combination, and we can, in fact view it as a function. That is what we saw in a few slides

ago, and we can also think of it as matrix multiplication. Just to remind you, we had

this thing called cA, which was the function, which was this linear combination 45x1 plus

125x2 plus 150x3 three, and which we can also write as a matrix multiplication.

So similarly, for shops B and C, we can write the cost function cB and cC. So, the shop

is capital C and the cost is little c. And expressions and matrix forms are as below.

So, CB is the linear combination 40x1 plus 120x2 plus 170x3 and that is the corresponding

matrix multiplication. And for shop C, we have the function CC, which is 50x1 plus 130x2

plus 160x3 and the corresponding matrix multiplication.

So, to compare these expressions, let us go back for a second and check them out. We had

45x1 plus 125x2 plus 150x3, 40x1 plus 120x2 plus 170x3. So, rice and dal are cheaper in

B than in a, but oil is more expensive. And then we have 50x1 plus 130x2 plus 160x3. So, rice and dal in C are more expensive than in shop B, but oil is less expensive.

However, when you compare it with shop A, all three are more expensive for shop C. So,

you would, assuming everything else is the same you would always prefer shop C, shop

A over shop C. So that is what we see here.

The third expression always yields larger values than the first one. Of course, we are

talking about commodities, so we think of them as buying the, I mean, buying commodities

from a grocery shop. So X1, X2, X3 are always I think of them as positive numbers, but we

do not know how to compare between the second shop that is shop B and shops A and C.

So, to compare this, we would want to compare these cost functions. So, the idea you would

I mean, the natural way of doing this would be to treat them together cA, cB and cC we

could think of them as a vector. So, you look at this vector, cA, cB, cC and then we can

think of this actually as a function. Because each of these, after all are functions from

R3 to R, so this vector can be thought of as an element of R3, it is a point of R3.

So, we can think of the cost function which is C of X1, X2, X3 is cA of X1, X2, X3 comma

cB of X1, X2, X3 comma cC of X1, X2, X3, I will just put this bracket here. And just

to recall each of these are given by linear expressions, linear combinations 45x1 plus

125x2 plus 150x3 and then 40x1 plus 120x2 plus 170x3 and then 50x1 plus 130x2 plus 160x3.

So, this is the function, the cost function, wherein

the first coordinate of the cost function tells you the cost in shop A, the second coordinate

tells you in shop B, and the third coordinate tells you in shop C. And now, when you are

given X1, X2, X3 you put it into this cost function, look at these three numbers that

it produces and then you choose the smallest one, you can just read them off. So, this

would be a good way of trying to compare. Of course, there are many other reasons to

look at the cost function.

For example, we can treat this as matrix multiplication. Because after all each of these are given

by matrix multiplication, where we have a column vector X1, X2, X3 on the right and

a row vector corresponding to the cost of individual items on the left. So, you can

put these three row vectors together and get this matrix.

So, C of X1, X2, X3 has a matrix form where the matrix is 45, 125, 150. This was the prices

for shop A, 40, 120, 170, which are the prices for shop B and for 50, 130, 160, which are

the costs for shop C. So, this is exactly what your table was the cost table, a few

slides ago. And you can see by just multiplying out that this is indeed what the cost vector

is.

So, just as an example, the cost of buying 2 kgs of rice, 1 kg of dal and 2 liters of

oil are given by the cost vector, you put X1 is 2, X2 is 1 and X3 is 2 and we will get

515, 540 and 550, if hopefully I have done this computation correctly.

So as in the case of the function cA, the property of linearity of costs can now be

extracted from the matrix form for the cost function c. So, let us first of all understand

this statement. What do we mean by the statement, the statement means that if you buy, so suppose

again, we had a situation like Monday and Tuesday, and then we had a say we wanted to

know what the costs for each shop on Wednesday are, so I would be evaluating this on half

times the rice on Monday, plus 5 by 4 times rice on Tuesday comma halftimes rice, dal

excuse me, dal on Monday plus 5 by 4 times dal on Tuesday, and then half times oil on

Monday plus 5 by 4 times oil on Tuesday.

You would be evaluating it on this, which means you have this matrix. Well, let me come

to the matrix in a second, which means basically you are doing cA of the same thing, cB of

the same thing and cC of the same thing, but we saw that cA is basically after we finished

the computations the cA of this is exactly half times cA on Monday. So, I will just call it Monday.

By Monday, what do I mean? I mean rice on Monday, the vector rice on Monday, dal on

Monday comma oil on Monday, plus 5 by 4 times cA on Tuesday, which means by T I mean the

vector rice on Tuesday, dal on Tuesday, oil on Tuesday. And then I can, the same way as

I did for cA the same. There is nothing special about the shop A. So, the same thing works

for cB as well and for cC as well.

So, I have this very nice expression, and then half times cC of Monday plus 5 by 4 times

cC on Tuesday. And so, I can effectively compute this by computing everything on Monday and

Tuesday, but it gets even better because we know how to do matrix. We know how to do,

how to add and subtract and scalar multiply things in R3.

So, because we can do that, we can write this in even better as half times the cost vector

evaluated on the commodities for Monday, plus 5 by 4 times the cost vector evaluated on the vector for Tuesday. I can separate these into two different vectors, and then I can

pull out the scalars, I will encourage you to do that. I will just, just in case it was

confusing what is M. By M I meant rice on Monday then dal on Monday, oil on Monday.

And by T I meant rice on Tuesday, dal on Tuesday, oil on Tuesday.

So, notice, I do not even remember what the quantity is there. I did not need that anywhere.

So, I can instead replace this rice and dal and so on by variables. I could have called

them X1, X2, X3 and Y1, Y2, Y3 and the whole thing would have worked out in the same way.

So effectively, what we are saying by linearity is that if you have alpha times X1, X2, X3

plus Y1, Y2, Y3 this is going to be just alpha times C of X1, X2, X3 plus C of Y1, Y2, Y3.

That is what we mean by linearity.

This is a very important property, and this is very special to the kind of functions that

we are dealing with, they are linear functions. So that is what this video was about. Remember, linear mapping. These are examples of linear mappings. Now, the claim is this can be, we

could have done this in one shot by using the matrix. How is that? Because remember

that, so let me just take here.

So, I can write this as the matrix times the vector X1, X alpha times X1, X2, X3 plus the

vector Y1, Y2, Y3. But we know that matrix multiplication commutes with the addition.

So, I can instead write this as the matrix, the cost matrix that we had in the earlier

slide, times X1, X2, X3, and I can pull the alpha out because of constants remember come

out plus the same matrix times Y1, Y2, Y3.

And then, of course, this I know is alpha times C of X1, X2, X3. I mean it is the same

as the expression on top. It is equal to this expression. This this part is here and the

second part is corresponding to C of Y1, Y2, Y3. So, this is what we mean by a linearity.

So now, we can talk in general, what is a linear mapping? So, a linear mapping f from

Rn to Rm. And now, we are not doing R3 or R we will just do it for any arbitrary n and

m. mean something of this form where you have f of X1, X2, Xn is summation a1jxj summation

a2jxj summation amjxj, what are these aijs, they are real numbers.

So, basically a linear mapping can be thought of as a combination or a, sorry, a collection

of linear combinations. Except you arrange them in a vector, so it gives you a function.

So linear mapping is a function, which has this form. So just to be clear, if you take

something like f of x is x squared, that is not a linear mapping or f of x is logarithm

of x that is not a linear mapping or f of x is e to the power x that is not a linear

mapping. So, it should be linear, meaning it has this is very nice form.

So, we can of course write this as matrix multiplication, because notice that the right-hand

side, if we think of it, instead of a row vector, if we think of it as a column vector

then we can obtain it as multiplying A times x. What is A? A is exactly the matrix given by putting these coefficients into the appropriate places. So, the matrix the ijth entry of A

is aij where aijs is the corresponding coefficient above.

So finally, let us talk about linearity of linear mappings. This is exactly what we did

for the cost vector. So, it follows that a linear mapping satisfies linearity, which

means f of X1 plus Cy1 comma X2 plus Cy2, comma Xn plus Cyn is equal to f of X1, X2,

Xn plus C times f of Y1, Y2, Yn.

Why is that? So, the reason is because I can write f of X1 plus Cy1 comma X2 plus Cy2,

Xn plus Cyn as A times X1 plus Cy1, X2 plus Cy2, Xn plus Cyn. So, here of course, I am,

when I write it as A times just, I get a column letter, but you think of it as a row vector.

So, when I say equal, I mean, you take the row vector corresponding to this column vector.

So, the second I mean the column vector over there X1 plus Cy1 I can break up into two

separate vectors along with a constant. And then because matrix multiplication is commutes with or distributes over addition, we get this, which is exactly f of X1, X2, Xn plus

C times f of Y1, Y2, Yn. You could do this directly also, by the way, and I will encourage

you to check this from the expression for the linear combinations.

So, just to recall what everything that we have done in this video. We saw the examples

of grocery shops, first of a single grocery shop and the corresponding cost vector, and

then of a bunch of grocery shops. And we saw that it was useful to arrange this as a function.

We can treat the cost as a function, it is a linear combination, and that is the expression

for the function, and it satisfies what is called linearity.

And then if we have several costs, which is what we had four different jobs, we arrange

them into a cost function of which is a function of three variables. Taking values, depending

on how many shops there are. So, it could be in, so we had three shops, so it took values

in R3 so it is a function from R3 to R3, but, each coordinate of the expression that we had was a linear combination of the X1, X2, Xn of X1, X2, X3. So, such a thing is exactly

what is called a linear mapping. And then we observed that linear mappings are satisfied

linearity. Thank you.